Quiz 5 - DSC 10, Fall 2025

Full Name:

PID:

Quiz Time: ◯ 2PM ◯ 3PM ◯ 4PM ◯ 5PM

Instructions:

- This quiz consists of 2 questions. You have a total of 20 minutes to complete it.
- Please write **clearly** in the provided answer boxes; we will not grade work that appears elsewhere. Completely fill in bubbles and square boxes; if we cannot tell which option(s) you selected, you may lose points.

  ◯ A bubble means that you should only **select one choice**.

  ☐ A square box means you should **select all that apply**.
- If your answer is a string, make sure to put it in quotes. If your answer is a float, make sure to include a decimal point.
- You may use one handwritten sheet of notes. No calculators and no computers.
- Assume we have already run `import babypandas as bpd` and `import numpy as np`.

By signing below, you are agreeing that you will behave honestly and fairly during and after this quiz.

Signature:

Version A

Please do not open your quiz until instructed to do so.

Leander collected a simple random sample of 500 Major League Baseball players, stored in the DataFrame `mlb` shown on the left below. The data includes each player's name (`"player"`), overall skill on a 0 to 100 scale (`"skill"`), `"team"`, `"position"`, and batting side (`"bat"`), which has values `"Left"` and `"Right"`. Another DataFrame, `soxcards` (shown on the right below), was created to summarize the distribution of positions for two teams, the Red Sox and the Cardinals, where `"IF"` represents all infield positions, including `"C"`, `"1B"`, `"2B"`, `"3B"`, and `"SS"`.

	player	skill	team	position	bat
0	David Ortiz	95	Red Sox	1B	Left
1	Shohei Ohtani	99	Dodgers	P	Left
2	Aaron Judge	97	Yankees	OF	Right
3	Corey Seager	90	Rangers	SS	Left
4	George Springer	88	Blue Jays	OF	Right

position	Red Sox	Cardinals
IF	0.37	0.41
OF	0.16	0.23
P	0.47	0.36

Question 1

a) Calculate the TVD between the two distributions in the `soxcards` DataFrame on the right.

b) Suppose we were to split the `"IF"` group into all of its individual positions and their respective proportions. How might the TVD change? Select all that **could** be true.

☐ increase ☐ decrease ☐ stay the same

Question 2

Leander conducts a hypothesis test to determine if there is evidence that players whose batting side is `"Left"` have higher `"skill"` values than those whose batting side is `"Right"`.

a) Fill in the blanks to complete the statements of the null and alternative hypotheses.

Null: The mean `"skill"` of `"Left"` batters is ___(a)___ the mean `"skill"` of `"Right"` batters.

Alt: The mean `"skill"` of `"Left"` batters is ___(b)___ the mean `"skill"` of `"Right"` batters.

(a): (b):

b) **Select all** of the possible test statistics that we could use for this hypothesis test.

☐ The absolute difference in mean `"skill"` between `"Left"` and `"Right"` batters.
☐ `"Left"` batters' maximum `"skill"` minus `"Right"` batters' maximum `"skill"`.
☐ The TVD between the `"skill"` distributions for `"Left"` and `"Right"` batters.
☐ None of the above.

c) Fill in the blanks in the code below to perform a permutation test with an appropriate test statistic and compute the p-value.

```
def get_test_stat(df, column):
    means = df.__(a)__(column).mean().get(__(b)__)
    return means.loc["Left"] - means.loc["Right"]

test_stats = np.array([])
for i in np.arange(5000):
    shuffled_series = np.random.permutation(__(c)__)
    shuffled_df = mlb.assign(shuffled=shuffled_series)
    simulated_test_stat = get_test_stat(__(d)__)
    test_stats = np.append(test_stats, simulated_test_stat)

observed_test_stat = get_test_stat(mlb, "bat")
p_value = np.count_nonzero(__(e)__) / 5000
```

(a): (b):

(c): (d):

(e):

d) Our test above produces a p-value of 0.038. **Select all** of the statements below that are true.

- ☐ At the 0.01 significance level, we fail to reject the null hypothesis.
- ☐ Using a p-value cutoff of 0.01 results in a test that is 5 times more significant than the test with a p-value cutoff of 0.05.
- ☐ A p-value is the probability, if the null hypothesis is true, of observing data at least as extreme as what we actually observed.
- ☐ A p-value of 0.038 means there is a 3.8% chance the null hypothesis is true.
- ☐ If we used a p-value cutoff of 0.05 and tested many different pairs of hypotheses in which the null hypothesis was actually true, then we would incorrectly reject the null about 5% of the time.
- ☐ If we increase the number of repetitions to 10,000 the p-value should increase.
- ☐ If our alternative hypothesis were two-sided instead of one-sided, our p-value would be smaller.
- ☐ None of the above.

Reminders:

- **Write your PID** on the front page and on the top right corner of page 3.
- If your answer is a string, make sure to put it in **quotes**.
- If your answer is a float, make sure to include a **decimal point**.
- Stay in your seat until the quiz is over. No leaving early.